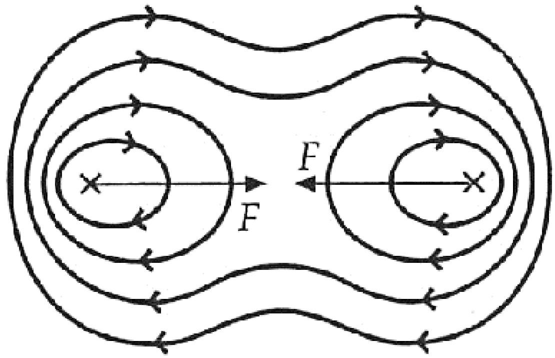
5	**(a)**		 **Correct direction of field lines** [B1] - as indicated by arrows on field lines **Correct shape and spacing of field lines** [B1] - field lines on outer sides of the currents have smaller spacing compared to field lines between the currents. Need to at least have two field lines on the outer sides to compare spacing. - symmetrical field pattern for two currents of same direction, with neutral point at the midpoint between P and Q. - field lines must be smooth, with no kinks and sharp edges.
	(b)		At the position of Q, the magnetic flux density due to current in P is directed out of the page. Since current in P is downwards, by Fleming's Left Hand Rule, [M1] The force on Q is towards the left. [A1]
	(c)	**(i)**	Vertically downwards [B1]
		(ii)	F = Bqv [C1] $6.4 \times 10^{-18} = B(1.6 \times 10^{-19})(4.0 \times 10^{6})$ [C1] $B = 1.0 \times 10^{-5}$ T [A1]
		(iii)	$\lvert B_{resultant} \rvert = \lvert B_Q \rvert - \lvert B_P \rvert$ Let I be the current in both P and Q $1.0 \times 10^{-5} = \dfrac{\mu_0 I}{2\pi(0.12)} - \dfrac{\mu_0 I}{2\pi(0.30 - 0.12)}$ [C1] $I = 18$ A [A1]

6	(a)		Faraday's Law of Electromagnetic Induction states that the induced e.m.f is